ANUJ KALE

Pune, Maharashtra | anujskale@gmail.com | Phone : +91 9168541230 | askyncme.github.io
Linkedin : www.linkedin.com/in/anuj-kale-672532291 | Github: https://github.com/askyncme

SUMMARY

Mechanical Engineering graduate with experience in autonomous UAVs, ROS2 robotics, computer vision, and embedded systems. Electronics Lead for Team Squadrone's SAE AeroTHON 2025 UAV project (AIR 11) and co-author of a published UAV research paper presented at DAUS 2026. Interested in navigation, guidance, control systems, and autonomous robotics.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **July 2022 - July 2026**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.12
- HSC : 77%
- SSC : 90.4%

Honors in Robotics & Automation

SKILLS

- **UAV & Robotics:** ROS2, Gazebo, RViz, PX4, ArduPilot, Mission Planner, UAV System Architecture
- **Programming:** Python, OpenCV, YOLO, Embedded C, Basic C++
- **Navigation & Perception:** Sensor Integration, Computer Vision, Path Planning Fundamentals
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi, Microcontroller Interfacing
- **Industrial Automation:** Mitsubishi FX3G PLC, Ladder Logic
- **Mechanical & Prototyping:** SolidWorks, 3D Printing

PROJECTS

1. **Team Squadrone - Autonomous UAV for Disaster Response**

SAE AeroTHON 2025 Phase I – AIR 11 | Research Paper Published at DAUS 2026, Salzburg, Austria

- Led electronics architecture and subsystem integration for an autonomous disaster-response UAV.
- Coordinated integration of flight controller, onboard computing, communication systems, sensors and mission payloads.
- Contributed to autonomous mission planning, target detection and payload deployment workflows using, Python and computer vision.
- Participated in UAV component selection, sensor integration, power system planning and simulation-based validation.
- Collaborated with multidisciplinary teams across electronics, mechanical and autonomous systems domains.
- Contributed to published research on terrain-aware target detection and autonomous payload delivery for UAV applications.

2. **Autonomous Mobile Assistant Robot (ROS2)**

- Developed a ROS2-based autonomous mobile robot using mecanum-drive kinematics.
- Implemented IMU and odometry integration for localization and navigation.
- Designed a navigation pipeline for autonomous indoor mobility.
- Utilized Gazebo and RViz for simulation, visualization and testing.

3. **Quadruped Robot**

- Led development of a 12-servo quadruped robot using Arduino Nano and 3D-printed components.
- Implemented gait control algorithms and servo coordination for stable locomotion.
- Conducted hardware integration, testing and performance validation.

4. **PLC-Based Pneumatic Gripper System**

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

- Published research paper at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.
- Investigated autonomous UAV systems for disaster-response missions, terrain-aware target detection and precision payload delivery.
- Explored integration of computer vision, onboard sensing and autonomous mission execution concepts.

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Supported technical procurement activities for warehouse robotics systems using SAP.
- Coordinated component sourcing, vendor communication and engineering support operations.
- Assisted in inventory tracking, documentation management and material planning.
- Gained exposure to robotics product-development workflows and cross-functional engineering operations.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi.
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.

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SUMMARY

Mechanical Engineering undergraduate with hands-on experience in robotics, automation, embedded systems, and prototype development. Worked on autonomous UAVs, PLC-based industrial automation, and mobile robotic systems involving electronics integration, testing, and system validation. Experienced in cross-functional project execution, troubleshooting, and rapid prototyping.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **Aug 2022**

Pune Vidyarthi Griha’s College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.01
- HSC : 77%
- SSC : 90.4%

Honors in Robotics & Automation

SKILLS

- **Robotics & Middleware:** ROS2, Gazebo, RViz, Robot system architecture
- **Programming:** Python, OpenCV
- **Embedded Systems:** Arduino, ESP-32, microcontroller interfacing
- **Industrial Automation:** Mitsubishi FX3G PLC
- **Mechanical & Prototyping:** SolidWorks, 3D Printing

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Supported procurement of components for warehouse robotics systems using SAP.
- Coordinated vendor communication and component tracking for engineering operations.

PROJECTS

Team Squadrone - Autonomous UAV for Disaster Response

Selected for presentation at DAUS 2026 – Salzburg, Austria | SAE AeroTHON 2025 Phase I – AIR 11

- Led electronics integration for an autonomous UAV, coordinating sensors, onboard computing, and flight controller systems.
- Contributed to vision-assisted detection and autonomous payload deployment; validated system via simulations and field tests.

Autonomous Mobile Assistant Robot (In Development)

- Designed ROS2-based mecanum-drive mobile robot with IMU and odometry-based localization.
- Implemented navigation pipeline for autonomous indoor movement.

Quadruped Robot

- Led development of a 12-servo quadruped using Arduino Nano with 3D-printed components.
- Implemented servo control logic and conducted gait testing.

PLC-Based Pneumatic Gripper System

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

Accepted at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.

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SUMMARY

Engineering undergraduate focused on robotics and embedded systems with hands-on experience in autonomous UAV development and industrial automation projects. Led electronics integration for a competition-grade drone and contributed to vision-assisted payload delivery research accepted at an international conference. Interested in practical autonomous system development and real-world deployment.

EDUCATION

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- CGPA : 8.01
- HSC : 77%
- SSC : 90.4%

Honors in Robotics & Automation

SKILLS

- **Robotics & Middleware:** ROS2, Gazebo, RViz, Robot system architecture
- **Programming:** Python, OpenCV&YOLO (Perception & Vision)
- **Embedded Systems:** Arduino,ESP-32, microcontroller interfacing
- **Industrial Automation:** Mitsubishi FX3G PLC
- **Mechanical & Prototyping:** SolidWorks, 3D Printing

PROJECTS

Team Squadrone - Autonomous UAV for Disaster Response
Selected for presentation at DAUS 2026 – Salzburg, Austria | SAE AeroTHON 2025 Phase I – AIR 11

- Led electronics integration for an autonomous UAV, coordinating sensors, onboard computing, and flight controller systems.
- Contributed to vision-assisted detection and autonomous payload deployment; validated system via simulations and field tests.

Autonomous Mobile Assistant Robot (In Development)

- Designed ROS2-based mecanum-drive mobile robot with IMU + odometry integration.
- Implemented navigation pipeline for autonomous indoor movement.

Quadruped Robot

- Led development of a 12-servo quadruped using Arduino Nano with 3D-printed components.
- Implemented servo control logic and conducted gait testing.

PLC-Based Pneumatic Gripper System

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

Accepted for presentation at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Supported procurement of components for warehouse robotics systems using SAP.
- Coordinated vendor communication and component tracking for engineering operations.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi, German(Basics).
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.

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EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **Aug 2022**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.01
- HSC : 77%
- SSC : 90.4%

Minors in Robotics & Automation

SKILLS

- **Programming :** Python (Basic), OpenCV & YOLO (Foundational)
- **Robotics & Middleware :** ROS2
- **Embedded Systems :** Arduino, Microcontroller Interfacing
- **Industrial Automation :** Mitsubishi FX3G PLC (Basic Ladder Logic)
- **Mechanical & Fabrication :** SolidWorks (Beginner), 3D Printing (Prototyping)

PROJECTS

Team Squadrone - Autonomous UAV for Disaster Response
Selected for presentation at DAUS 2026 – Salzburg, Austria | SAE AeroTHON 2025 Phase I – AIR 11

- Led electronics integration for an autonomous UAV, coordinating sensors, onboard computing, and flight controller systems.
- Contributed to vision-assisted detection and autonomous payload deployment; validated system via simulations and field tests.

Autonomous Mobile Assistant Robot (In Development)

- Designing a mecanum-wheel-based indoor robot for smart lab automation and mobility tasks.
- Developing system architecture for navigation, interaction workflows, and modular expansion.

Quadruped Robot

- Led development of a 12-servo quadruped using Arduino Nano with 3D-printed components.
- Implemented servo control logic and conducted gait testing.

PLC-Based Pneumatic Gripper System

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

Accepted for presentation at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Processed Purchase Orders using SAP-based systems to support technical procurement.
- Coordinated vendor communication and component tracking for engineering operations.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi, German(Basics).
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.

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SUMMARY

Engineering graduate focused on robotics, embedded systems, and autonomous platforms. Hands-on experience integrating sensors, microcontrollers, and control systems through academic projects and competitions. Contributed to team-based development of functional prototypes, with exposure to electronics integration, system-level debugging, and mechanical implementation. Interested in practical problem-solving and building reliable, real-world automation systems.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering — **Aug 2022 - Aug 2026**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.01
- HSC : 77%
- SSC : 90.4%

Minors in Robotics & Automation

SKILLS

- ROS2 Jazzy
- Basic Python - OpenCV
- SolidWorks
- 3D Printing
- Electronics integration

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune — **Dec 2024 - Jan 2025**

- Utilized SAP-based procurement software to execute procurement processes, including raising & generating Purchase Orders (PO).
- Gained practical exposure to the end-to-end procurement cycle, from material requirement identification to vendor coordination.
- Communicated with suppliers for quotations, order follow-ups, and material delivery timelines, enhancing understanding of procurement communications in a tech-driven environment.

PROJECTS

Electronics Lead for Team Squadrone - Autonomous Drone

- Secured 11th Rank in India (Phase 1) at AeroTHON 2025 among 50+ teams.
- Contributed to development of an autonomous drone for disaster missions.

Pneumatic Cylinder Gripper System

- Built a basic object-grabbing system using a pneumatic cylinder, a 5/3 double solenoid valve, push buttons, and sensors
- Controlled via Mitsubishi FX3G-14M PLC and ladder logic.

Quadruped Robot

- Led a team for developing a spider-like quadruped using Arduino Nano and 12 servos.
- Designed 3D printed leg components for flexibility and strength.

Mobile Assistant Robot for Smart Lab Environment

- Developing an autonomous mobile robot for smart lab environments using mecanum wheels, integration of AI modules (facial recognition, voice interaction) for lab management.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi, German(Basics).
- **Extracurricular Interests :** Held leadership roles in MESA (Media Head) and Team Squadrone (Electronics Head); led robotics projects; contributed to EDC Media and TEDx Design teams.
- **Achievements :** Ranked 11th out of 80+ teams in SAE AeroTHON Phase 1 (2025), NPTEL Product Design and Development, NPTEL Manufacturing Guidelines for Product Design.

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SUMMARY

Enthusiastic and hands-on engineering student with experience in robotics, embedded systems, and mechanical design. Skilled in leading team-based projects and collaborating across domains to build functional prototypes. Adept at applying technical knowledge to real-world challenges through internships, competitions, and club initiatives.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **Aug 2022 - Aug 2026**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- TE CGPA : 8.40
- HSC : 77%
- SSC : 90.4%

Honors Course in Robotics

SKILLS

- ROS2 Jazzy
- Basic Python - OpenCV
- SolidWorks
- 3D Printing
- Electronics integration

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Utilized SAP-based procurement software to execute procurement processes, including raising & generating Purchase Orders (PO).
- Gained practical exposure to the end-to-end procurement cycle, from material requirement identification to vendor coordination.
- Communicated with suppliers for quotations, order follow-ups, and material delivery timelines, enhancing understanding of procurement communications in a tech-driven environment.

PROJECTS

Electronics Lead for Team Squadrone - Autonomous Drone

- Secured 11th Rank in India (Phase 1) at AeroTHON 2025 among 50+ teams.
- Contributing to development of an autonomous drone for disaster missions.

Pneumatic Cylinder Gripper System

- Built a basic object-grabbing system using a pneumatic cylinder.
- Controlled via Mitsubishi FX3G-14M PLC and ladder logic.
- Integrated a 5/3 double solenoid valve, push buttons, and sensors.

Quadruped Robot

- Led a team for developing a spider-like quadruped using Arduino Nano and 12 servos.
- Designed 3D printed leg components for flexibility and strength.
- Controlled via Bluetooth; plans underway to enhance mobility and autonomy.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi, German(Basics).
- **Extracurricular Interests :** Held leadership roles in MESA (Media Head) and Team Squadrone (Electronics Head); led robotics projects; contributed to EDC Media and TEDx Design teams.
- **Achievements :** Ranked 11th out of 80+ teams in SAE AeroTHON Phase 1 (2025), NPTEL Product Design and Development, NPTEL Manufacturing Guidelines for Product Design.